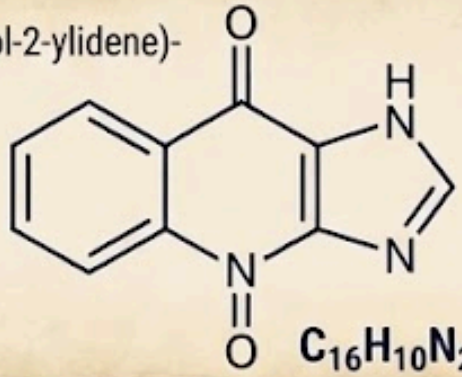
EXPLORING THE BLUES OF Indigo

IUPAC Name:
2-(1,3-dihydro-3-oxo-2H-indol-2-ylidene)-1,2-dihydro-3H-indol-3-one

Molecular Formula:
$C_{16}H_{10}N_2O_2$

Molar Mass:
262.26 g/mol



$C_{16}H_{10}N_2O_2$

Functional groups:
amide-like carbonyls, imide/nitrogen-containing heterocycle



Key component of **INDIGO VATS!** Responsible for the timeless blue hue.

1. Structure & Properties

- **Common name:** Indigo Dye
- **Family:** Indolyl (bis-indole derivative)
- **Appearance:** Deep blue solid/powder
- **Solubility:** Very low in water; soluble in organic solvents

2. Essential Properties

- Intense deep blue color
- Resistant to fading (light fastness)
- Reduction-oxidation active (vat dye)
- Forms as a reduction product of a precursor

3. What Makes it Special?

- Historic signifier of wealth Global trade driver
- Complex chemistry of 'vat dyeing'
- Distinctive, non-toxic organic dye

4. Creative & Industrial Uses

- Primary dye for denim & textiles
- Food colorant (under specific regulations)
- Intermediate in synthesis
- Limited use in inks and pigments

5. Synthesis & Natural Source

- Traditionally from Indigofera plant leaves
- Modern large-scale synthetic production
- Precursor indican to indigo conversion process

6. Environmental & Sustainable Chemistry

- Importance of wastewater treatment from dyeing
- Blo-indigo production research
- Opportunities for green reduction methods

Reliable Citations

1. National Center for Biotechnology Information (NCBI). (2024). *PubChem Compound Summary for CID 10212, Indigo*. Bethesda (MD): National Library of Medicine (US).
2. Bechtold, T., & Mussak, R. (Eds.). (2009). *Handbook of Natural Colorants* (pp. 125-144). West Sussex, UK: John Wiley & Sons, Ltd. (Specifically the chapter on Indigo

Safety & Ethics Note

SAFETY NOTE: Dye powder can be an irritant. Handle with care.
ETHICS: Note historical context